

Unique decomposition Theorem.

For any $B \in GL_n(\mathbb{C})$, there exists a unique lower-triangular M with positive diagonal s.t. MB is unitary.

Proof. Since B is invertible, the row vectors of B , $\{\beta_1, \dots, \beta_n\}$ is basis for V .

Apply Gram-Schmidt process, that is, for each $1 \leq k \leq n$, put

$$\alpha_k = \beta_k - \sum_{j < k} \frac{\langle \beta_k, \alpha_j \rangle}{\|\alpha_j\|^2} \alpha_j$$

Then, $\text{span}\{\alpha_1, \dots, \alpha_r\} = \text{span}\{\beta_1, \dots, \beta_r\}$ and $\{\alpha_1, \dots, \alpha_r\}$ ($r \leq n$) is orthogonal.

Put $U = \begin{bmatrix} \frac{\alpha_1}{\|\alpha_1\|} & \dots & \frac{\alpha_n}{\|\alpha_n\|} \end{bmatrix}$, then clearly $U^* U = I$, because $\{\alpha_1, \dots, \alpha_n\}$ is orthogonal.

And, for each $1 \leq k \leq n$, $\alpha_k = \beta_k - \sum_{j < k} C_{kj} \beta_j$ for some C_{kj} .

Put $M_{kj} = \begin{cases} -\frac{C_{kj}}{\|\alpha_k\|} & \text{if } k > j \\ \frac{1}{\|\alpha_k\|} & \text{if } k = j \\ 0 & \text{otherwise.} \end{cases}$ Then, each $1 \leq k \leq n$, $U_k =$

$$\begin{pmatrix} & & 0 \\ & & \\ M_{k3} & & \\ & & \\ & & \end{pmatrix} \cdot (\beta_1 \dots \beta_n) = U$$

so $MB = U$. To show the uniqueness, denote

$$T^+(n) = \{M \in M_{nn}(\mathbb{C}) \mid M \text{ is lower triangular, positive diagonals}\}.$$

Let $M_1, M_2 \in T^+(n)$ satisfies $M_1 B, M_2 B$ are unitary. Since unitary matrices forms a group

$$(M_1 B) \cdot (M_2 B)^{-1} = M_1 M_2^{-1} \text{ is also unitary.}$$

And, M_2^{-1} is lower-triangular with positive diagonals, so is $M_1 M_2^{-1}$. Now,

$$(M_1 M_2^{-1})^{-1} = (M_1 M_2^{-1})^*, \text{ so } M_1 M_2^{-1} \text{ is diagonal.}$$

So, $M_1 M_2^{-1} = I$, or $M_1 = M_2$. $\square$

Corollary. For each $B \in GL(n)$, there exists unique matrices K, A, N s.t. $B = KAN$, where K is unitary, A is diagonal with positive values, N is upper-triangular with diagonals 1

+Note. Let $T_n(F)$ be the set of $n \times n$ upper-triangular matrices with all diagonals are 1.

Then, $(T_n(F), \cdot)$ is group, because: for any $A \in T_n(F)$, $A = I + N$,

where N is strictly upper triangular with $N^n = 0$, nilpotent. So,

$$A^{-1} = I - N + N^2 - N^3 + \dots + (-N)^{n-1} \left(= \sum_{k=0}^{n-1} (-N)^k \right)$$

and $A^{-1} \in T_n(F)$ is clear.

QR- Factorization.

Let $A = (w_1 \dots w_n)$ be an invertible matrix.

And, for each $1 \leq k \leq n$,
$$V_k = w_k - \sum_{i=1}^{k-1} \frac{\langle w_k, V_i \rangle}{\|V_i\|^2} \cdot V_i.$$

Then, $\{V_1, \dots, V_n\}$ is an orthogonal, so put $U_k = V_k / \|V_k\|$, then

$\{U_1, \dots, U_n\}$ is an orthonormal. Clearly, $Q \stackrel{\text{def}}{=} (U_1 \dots U_n)$ is Unitary.

Moreover, for each $1 \leq k \leq n$,

$$w_k = \|V_k\| \cdot U_k + \sum_{i=1}^{k-1} \langle w_k, U_i \rangle \cdot U_i. \left(= V_k + \sum_{i=1}^{k-1} \frac{\langle w_k, V_i \rangle}{\|V_i\|^2} \cdot V_i \right)$$

Define

$$R_{jk} = \begin{cases} \|V_k\| & j=k \\ \langle w_k, U_j \rangle & j < k \\ 0 & j > k \end{cases}$$

Then, R is upper-triangular, and satisfying

$$Q^{-1}A = Q^* A = \begin{pmatrix} \overline{U_1} \\ \vdots \\ \overline{U_n} \end{pmatrix} \cdot (w_1 \dots w_n) = \begin{pmatrix} \langle w_1, U_1 \rangle & \langle w_2, U_1 \rangle & \dots & \langle w_n, U_1 \rangle \\ \langle w_1, U_2 \rangle & & & \\ \vdots & & & \\ \langle w_1, U_n \rangle & & & \langle w_n, U_n \rangle \end{pmatrix} \\ = R.$$

so, $A = QR = QDR'$

$$\text{where } D = \begin{pmatrix} \|V_1\| & & 0 \\ & \|V_2\| & \\ 0 & & \ddots \\ & & & \|V_n\| \end{pmatrix} \quad \text{and} \quad R' = \begin{pmatrix} 1 & & & \\ & 1 & \frac{\langle w_k, U_j \rangle}{\|V_j\|} & \\ & & \ddots & \\ 0 & & & 1 \end{pmatrix}$$